How Accurate Is Regression-Based Global Effect Estimation Under Network Interference? Leverage, Extrapolation and Misspecification

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Abstract

Regression-based estimators of the global average treatment effect (GATE) under network interference fit outcomes on a unit's treatment and neighbourhood exposure, then predict outcomes with everyone and with no one treated. How accurate can they be, and what determines their error? We decompose the error into a design-leverage term and an extrapolation-bias term and derive closed forms for the one-hop fraction regression: $\mathrm{RMSE}^2 \approx B^2 + \kappa L/n$, with $L = (1+d_H)/(p(1-p))$ under unit randomization (d_H the harmonic-mean degree) and a cluster-design analogue, plus bias formulas for nonlinear exposure response (the regression recovers the response slope at p, not the secant), omitted multi-hop spillover ($\approx -\beta\lambda^2/(1-\lambda)$) and degree-related heterogeneity (a $1/d_i$-weighted estimand). In simulations on synthetic graphs and two Facebook graphs (Stanford3, ego-Facebook) with exactly known GATE, one fitted constant predicts the RMSE of 162 Bernoulli-design runs (median ratio 1.01, 98% within 10%) and the bias formulas match simulation to within about 0.01–0.02 on a GATE of 2. Biases do not shrink with n; cluster randomization helps in proportion to how well the graph clusters; richer regressions help only where the graph supplies units with extreme exposures; regression beats difference-in-means only when spillover exceeds $\approx \hat{\kappa}\sqrt{L/n}$; sandwich intervals cover well when the model is right (mean 0.95) and fail when it is not (0.57). Outcomes are synthetic, and the closed forms are population-level and approximate for cluster designs.

1 Introduction

Online A/B tests on social networks and marketplaces violate the no-interference assumption: a unit's outcome depends on the treatments of its neighbours. The quantity a platform usually wants is the *global average treatment effect* (GATE), $\tau = \frac{1}{n}\sum_i [Y_i(\mathbf{1}) - Y_i(\mathbf{0})]$, the difference between mean outcomes if everybody and if nobody were treated. The usual difference in means (DiM) is biased for it. Inverse-probability (Horvitz–Thompson, Hájek) estimators of exposure-based contrasts [Aronow and Samii, 2017, Ugander et al., 2013] are unbiased for the GATE under a correct exposure model, but need units whose whole neighbourhood is treated (or untreated); on dense graphs such units barely exist and variance explodes. *Regression-based* estimators [Chin, 2019, Han and Ugander, 2023] sidestep this: fit y_i on the unit's treatment and graph-derived exposure features from the experimental data, then predict outcomes under the all-treated and all-control assignments and average the difference. They use every unit and are cheap, but how accurate they can be, what determines their error, and when they fail is only partly understood.

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This paper. We study these questions with controlled simulations on synthetic graphs (Erdős–Rényi, stochastic block model, Watts–Strogatz, Barabási–Albert, random regular) and two public Facebook graphs (Stanford3, $n = 11{,}586$; ego-Facebook, $n = 4{,}039$), with synthetic outcomes so that the true τ is known exactly. The device is a decomposition (Sec. 4) of the error into a *design-leverage* term, set by how far the all-treated/all-control exposures lie from those the design generates, and a *bias* term set by misspecification, with closed forms for the one-hop fraction-of-treated-neighbours regression. Findings:

1. **Sampling error is predictable from graph and design alone.** $\mathrm{RMSE} \approx \hat{\kappa}\sqrt{L/n}$ with leverage $L = (1 + d_H)/(p(1-p))$ under unit randomization (d_H: harmonic-mean degree): error grows like $\sqrt{\text{degree}}$, shrinks like $n^{-1/2}$, and one fitted constant predicts 162 Bernoulli-design runs with median ratio 1.01 (98% within $\pm 10\%$). Cluster randomization lowers L by an amount that depends on how well the graph clusters (RMSE $2.9\times$ lower on ego-Facebook, $1.9\times$ on Stanford3); the closed form is a rough guide there.
2. **Four bias mechanisms with closed-form predictions that match simulation:** extrapolation under nonlinear exposure response (the regression recovers the slope at p, not the secant; Prop. 3); omitted multi-hop spillover ($B \approx -\beta\lambda^2/(1-\lambda)$; Prop. 5); effect heterogeneity (OLS targets a $1/d_i$-weighted average of unit effects; Prop. 4; bias up to 100% of the GATE on Stanford3); exposure mis-definition. None shrinks with n; predicted bias and leverage jointly match the RMSE of 360 Bernoulli world-settings (median ratio 1.005).
3. **Design matters as much as the estimator.** Clustering cuts leverage and, where the graph clusters well, bias (multi-hop bias $-0.71 \to -0.11$ on ego-Facebook, $-0.85 \to -0.49$ on Stanford3, little on ER/SBM); the treated fraction and number of clusters trade bias for variance.
4. **Rules and negative results.** Regression beats DiM only if spillover exceeds $\approx \hat{\kappa}\sqrt{L/n}$ ($\geq 0.8\beta$ for $n{=}10^3$, $d{=}128$); Hájek estimators are undefined on dense random graphs ($d \geq 32$) under unit randomization; richer regressions reduce bias only where the graph supplies extreme exposures (a cubic fixes Stanford3/ego-Facebook for 15–45% extra RMSE but has RMSE ≈ 2 on near-regular graphs); sandwich intervals cover well when the model is right (mean 0.95) and fail when it is not (mean 0.57), the failure being explained by the bias.

2 Related work

Designs and exposure-based estimation. Aronow and Samii [2017] give randomization-based inverse-probability estimators for general exposure mappings; graph cluster randomization [Ugander et al., 2013], its randomized variant [Ugander and Yin, 2023] and rate-optimal cluster designs [Leung, 2022] reduce the variance of such estimators; Eckles et al. [2017] study interference bias and neighbourhood-aware analysis; Karrer et al. [2021] describe a deployed system; Chen et al. [2023] optimize assignment covariances for HT estimators. **Outcome models.** Cortez et al. [2022], Cortez-Rodriguez et al. [2023] and Eichhorn et al. [2024] obtain total-effect estimators under low-order interaction models (the last with clustered designs); Yu et al. [2022] work without the network. Chin [2019] proposes the regression-adjustment estimator studied here; Han and Ugander [2023] select graph- and treatment-based covariates with block bootstrap. Sävje et al. [2021] and Li and Wager [2022] give asymptotics under limited interference and random graphs. **This work** asks a diagnostic question about the regression estimator family: what determines its error, and which factor dominates when. Our contributions are closed-form leverage and bias predictors validated against simulation on real graph topologies, a head-to-head with difference-in-means and Hájek baselines, and an audit of interval coverage. We propose no new estimator; the weighted and stratified fits in Sec. 6 are diagnostics-motivated fixes whose merits we only evaluate.

3 Setting, estimators and metrics

Setting. A connected undirected graph with adjacency A on n units, degrees $d_i \geq 1$, $W = D^{-1}A$. Assignment $z \in \{0,1\}^n$ is drawn either by *Bernoulli*(p) (independent units) or by *cluster randomization*: units are partitioned into clusters (Louvain communities [Blondel et al., 2008], resolution tuned to a target number K of clusters) and clusters are independently treated with probability p (redrawing the rare assignments in which one arm is empty). Potential outcomes $Y_i(z)$ are fixed given a "world" (graph plus unit-level parameters); the only randomness within a

world is z. Exposure features are $s_i = (Wz)_i$ (fraction of treated neighbours), $c_i = (Az)_i$ (count), $s_i^{(2)} = (W^2 z)_i$ (two-step fraction).

Regression estimators. $\hat{\tau} = \frac{1}{n}\sum_i [\hat{f}(x_i(\mathbf{1})) - \hat{f}(x_i(\mathbf{0}))]$ where $x_i(z)$ are the features and $\hat{f}$ is fitted on the observed data. Linear models: $(1, z, s)$ [OLS-S, the default, Chin-type]; $(1, z, c)$ [OLS-CNT]; $+zs$ [INT]; $+s^2$ [QUAD]; $+s^2, s^3$ [CUB]; $+s^{(2)}$ [OLS-S2]; $+s \cdot \ell_i,\ \ell_i$ (standardized log degree) [S-DEG]; weighted least squares with weights $\propto 1/\mathrm{Var}(s_i)$ [WLS-S]; separate OLS-S fits in five degree strata, averaged with stratum sizes [STRAT-S]; gradient boosting (scikit-learn HistGradientBoosting) on $(z, s, s^{(2)}, \ell)$ [GBM]. **Baselines:** DiM; Hájek estimator with exposure "unit and all neighbours treated" / "none treated" with exactly computed design probabilities (undefined when no unit qualifies in an arm).

Metrics. For each setting we report, over $R = 300$ assignments in each of 6 independent worlds, the *bias* $\mathbb{E}\hat{\tau} - \tau$, *SD*, and *RMSE*$= \sqrt{\mathbb{E}(\hat{\tau} - \tau)^2}$; RMSE is our headline metric because it combines bias and variance, which have different drivers; we show both. Tables and figures give means across worlds with min–max across the 6 worlds as error bars (Monte-Carlo standard errors of each world's RMSE are $\leq 0.04\times$RMSE with $R = 300$ and are smaller than the across-world spread). For inference we report empirical coverage of nominal 95% intervals $\hat{\tau} \pm 1.96\,\widehat{\mathrm{se}}$.

4 What determines the error: a decomposition and closed-form predictors

Throughout, the randomness is the assignment z only (design-based); potential outcomes are fixed. Let $\phi_i(z) \in \mathbb{R}^k$ be the feature map, $\hat{\Sigma} = \frac{1}{n}\sum_i \phi_i \phi_i^\top$ and $\hat{\beta} = \hat{\Sigma}^{-1}\frac{1}{n}\sum_i \phi_i y_i$. The estimator is $\hat{\tau} = \bar{d}^\top \hat{\beta}$ with $\bar{d} = \frac{1}{n}\sum_i [\phi_i(\mathbf{1}) - \phi_i(\mathbf{0})]$ (for $\phi = (1, z, s)$: $\bar{d} = (0, 1, 1)$). Let $\Sigma = \mathbb{E}\hat{\Sigma}$, $\beta^\star = \Sigma^{-1}\frac{1}{n}\sum_i \mathbb{E}[\phi_i y_i]$ be the population least-squares coefficient under the design, $\varepsilon_i = y_i - \phi_i^\top \beta^\star$ and $\hat{g} = \frac{1}{n}\sum_i \phi_i \varepsilon_i$ (so $\mathbb{E}\hat{g} = 0$).

Proposition 1 (Decomposition)**.** *Exactly, $\hat{\tau} - \tau = B + \bar{d}^\top \hat{\Sigma}^{-1}\hat{g}$ with the* extrapolation bias $B = \bar{d}^\top \beta^\star - \tau$*, which does not depend on n through any sampling effect. To first order ($\hat{\Sigma} \to \Sigma$) the second term has mean zero and variance at most $\kappa L/n$, where $L = \bar{d}^\top \Sigma^{-1}\bar{d}$ is the* design leverage *of the all-treated-vs-all-control contrast and $\kappa = \lambda_{\max}\big(\Sigma^{-1/2}\, n\mathrm{Cov}(\hat{g})\, \Sigma^{-1/2}\big)$ is the effective residual scale ($\kappa \approx \sigma^2$ for homoskedastic independent residuals). Hence* $\mathrm{RMSE}^2 \approx B^2 + \kappa L/n$.

L depends on the graph, design and feature map but not on the outcomes: it is the price of extrapolating from the exposures the design generates to the corners "everyone treated" and "no one treated". For $\phi = (1, z, s)$ (Chin-type estimator) it has a closed form.

Proposition 2 (Leverage of the one-hop fraction model)**.** *Let $v = p(1-p)$, $w_{ic} = m_{ic}/d_i$ where m_{ic} is the number of neighbours of i in cluster c, $\rho = \frac{1}{n}\sum_i w_{i,c(i)}$ and $q = \frac{1}{n}\sum_i \sum_c w_{ic}^2$. Under independent cluster-level Bernoulli(p) assignment (unit-level Bernoulli is the case of singleton clusters, $\rho = 0$, $q = 1/d_H$ with $d_H = (\frac{1}{n}\sum_i 1/d_i)^{-1}$ the harmonic-mean degree),*

$$L = \frac{1}{v}\Big[1 + \frac{(1-\rho)^2}{q - \rho^2}\Big], \qquad \textit{in particular} \qquad L_{\mathrm{Bern}} = \frac{1 + d_H}{p(1-p)}.$$

The interference penalty over the no-interference variance $1/v$ is $(1-\rho)^2/(q-\rho^2)$: d_H for unit randomization, $\to 0$ when all neighbours share the unit's cluster ($\rho \to 1$; z and s collinear, as precise as DiM). So $\mathrm{RMSE} \propto \sqrt{(1 + d_H)/(vn)}$ under unit randomization.

Proposition 3 (Extrapolation bias under a nonlinear exposure response)**.** *Let $Y_i(z) = a_i + \beta_i z_i + \gamma\, g(s_i)$ with $g(0) = 0,\ g(1) = 1$, and Bernoulli(p) assignment (heterogeneous γ_i with mean γ, independent of degree, give approximately the same B). For $\phi = (1, z, s)$,*

$$B = \gamma\,(\kappa_p - 1), \qquad \kappa_p = \frac{\sum_i \mathrm{Cov}(g(s_i), s_i)}{\sum_i \mathrm{Var}(s_i)}, \quad s_i \sim \mathrm{Bin}(d_i, p)/d_i,$$

and $\kappa_p \to g'(p)$ when all degrees grow, so that $B \to \gamma\,(g'(p) - 1)$: the regression recovers the slope of the response at p, not the secant slope between 0 and 1.

For $g(s) = s^k$ at $p = \frac{1}{2}$, $g'(p) = k2^{1-k}$: $B = 0$ for $k = 2$ (tangent at $\frac{1}{2}$ equals secant), $-\gamma/2$ for $k = 4$, $\approx -0.29\gamma$ for $k = \frac{1}{2}$; steep thresholds centred at p give large positive bias. B is not reduced by more data, nor (Sec. 6) by more flexible regressions when, under unit randomization with large degree, essentially no unit has exposure near 0 or 1.

Proposition 4 (Heterogeneous effects: the estimand is variance-weighted)**.** *Let $Y_i(z) = a_i + \beta_i z_i + \gamma_i s_i$ and Bernoulli assignment. For $\phi = (1, z, s)$, $\beta^\star$ yields $\bar{d}^\top \beta^\star = \bar{\beta} + \sum_i \pi_i \gamma_i$, $\pi_i = d_i^{-1} / \sum_j d_j^{-1}$, whereas $\tau = \bar{\beta} + \bar{\gamma}$ (uniform weights). The bias is $B = \sum_i (\pi_i - \frac{1}{n})\gamma_i$.*

OLS weights unit effects by the variance $\propto 1/d_i$ of the unit's exposure, so low-degree units dominate; if susceptibility correlates with degree the bias can be of the order of the effect itself, whatever n.

Proposition 5 (Omitted multi-hop spillover)**.** *Let $Y(z) = (I - \lambda W)^{-1}(a + \beta z)$, $W = D^{-1}A$, $|\lambda| < 1$, with Bernoulli assignment and $\phi = (1, z, s)$. Define $r_k = \frac{1}{n}\mathrm{tr}\, W^k$, $u_k = \frac{1}{n}\sum_{ij}(W^k)_{ij} W_{ij}$, $q = u_1$ and $\psi_k = r_k + u_k/q$. Then*

$$B = -\beta \sum_{k \geq 2} \lambda^k (1 - \psi_k) \;\approx\; -\beta \, \frac{\lambda^2}{1 - \lambda} \quad (\psi_k \approx 0).$$

ψ_k is the share of k-step influence aliased with the included one-hop terms (return and closure probabilities); on tree-like graphs the regression misses all indirect effect beyond one hop. Under cluster randomization distant exposures correlate with s and are partly absorbed (Sec. 6); we give no closed form for that case.

Proofs (Appendix A) are design-expectation statements; finite-n and finite-cluster effects are quantified empirically.

5 Experimental protocol

Graphs. Synthetic ($n = 4000$ unless stated): Erdős–Rényi (ER) and random regular with mean degree $d \in \{4, \ldots, 256\}$; stochastic block model (SBM, 40 equal blocks, $p_{\mathrm{in}}/p_{\mathrm{out}} = 30$); Watts–Strogatz (WS, rewiring 0.05; strong local clustering); Barabási–Albert (BA, $m \in \{3, \ldots, 24\}$; heavy-tailed degrees). Real: Stanford3 (Facebook100 [Traud et al., 2012]; Network Repository [Rossi and Ahmed, 2015]; $n = 11{,}586$, 568,309 edges, mean/median/max degree $98/73/1172$, mean local clustering 0.24) and ego-Facebook (SNAP [McAuley and Leskovec, 2012]; $n = 4{,}039$, 88,234 edges, mean/median/max degree $44/25/1045$, clustering 0.61). Both are connected. Data versions and checksums are in Appendix C. Clusterability differs sharply: the Louvain partition into ≈ 30 clusters cuts 18% of ego-Facebook's edges but 58% of Stanford3's.

Potential outcomes. Unit-level draws are made once per world: baseline $a_i \sim N(0, 1)$, direct effect $\beta_i = 1 + 0.5\xi_i$, spillover coefficient $\gamma_i = \gamma(1 + 0.5\zeta_i)$, $\xi, \zeta \sim N(0, 1)$ (so $\bar{\beta} \approx 1$, $\bar{\gamma} \approx \gamma$, default $\gamma = 1$). There is no extra observation noise; baselines and effect heterogeneity act as design-based noise. Mechanisms: **(i) one-hop fraction**, $Y_i = a_i + \beta_i z_i + \gamma_i g(s_i)$ with $g(s) = s^k$ (linear $k = 1$; concave $k \in \{\frac{1}{4}, \frac{1}{2}\}$; convex $k \in \{2, 4\}$) or a threshold response $g(s) \propto \sigma((s - \frac{1}{2})/0.08)$ rescaled to $g(0) = 0, g(1) = 1$ so that $\tau \approx \bar{\beta} + \bar{\gamma}$ for every shape; **(ii) multi-hop** (linear-in-means equilibrium) $Y = (I - \lambda W)^{-1}(a + \beta z)$, $\lambda \in [0, 0.8]$, $\tau = \bar{\beta}/(1 - \lambda)$ in expectation; **(iii) count**: $Y_i = a_i + \beta_i z_i + \gamma_i c_i/\bar{d}$; **(iv) degree-heterogeneous** spillover: (i) with $\gamma_i \propto e^{\mp 0.7\ell_i}$ (ℓ_i standardized log degree; "inv": low-degree units more susceptible; "pos": high-degree units), normalized to mean γ. The true τ is computed exactly from $Y(\mathbf{1}) - Y(\mathbf{0})$.

Experiments and worlds. Seven experiments (E1–E7; scripts in Appendix C) vary degree, n and noise (E1); response shape (E2); neighbourhood, heterogeneity and exposure definition (E3); spillover strength (E4); treated fraction and number of clusters (E5); inference (E6); a mis-measured graph (E7). Each setting uses 6 independent *worlds* (a fresh synthetic graph and unit parameters, or fresh unit parameters on the fixed real graph) with $R = 300$ assignments each (400 in E6, 40 for GBM). Hypotheses H1–H6 (error $\propto$ leverage; persistent extrapolation bias; multi-hop omission bias; heterogeneity bias; clustering helps iff the graph clusters; regression beats DiM only above a spillover threshold) were fixed before the final runs. A three-world pilot of E1/E3 set the run scale and exposed a crash (assignments with an empty arm under few clusters), fixed in the released code; WLS-S and STRAT-S were added after the pilot heterogeneity results and are labelled as such.

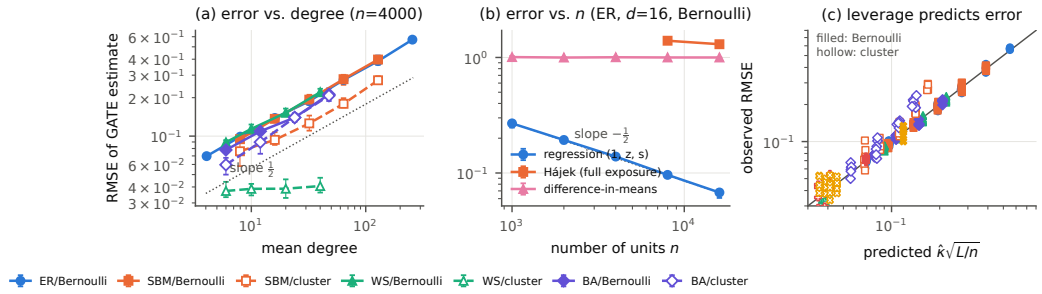


Figure 1: Variance side (E1). (a) RMSE of OLS-S vs mean degree for five graph families under Bernoulli (solid) and cluster randomization (dashed, K=40 target clusters); dotted: slope $\frac{1}{2}$. (b) RMSE vs n on ER graphs with d=16: regression, Hájek (shown only where defined in $> 50\%$ of assignments) and DiM. (c) Observed RMSE vs $\hat{\kappa}\sqrt{L/n}$ (closed-form leverage; one global $\hat{\kappa}$ fitted on ER/Bernoulli runs) for 288 runs including Stanford3 and ego-Facebook. Markers: mean over 6 worlds; bars: min–max over worlds (panels a–b); panel c shows each setting-world pair mean.

6 Results

6.1 Sampling error is set by design leverage (H1)

With a correctly specified linear one-hop response, the RMSE of OLS-S grows with mean degree with log-log slope 0.52 in $\log(1+d)$ (ER) and falls with n with slope -0.50 (Fig. 1a–b), as Prop. 2 implies. DiM does not improve with n (its error is bias); Hájek is defined in more than half of the draws only for $n \geq 8000$ at $d = 16$ and has RMSE 1.3–1.4 there. With one constant $\hat{\kappa} = 1.09$ fitted on the 72 ER/Bernoulli runs, $\hat{\kappa}\sqrt{L/n}$ matches the observed RMSE of all 162 Bernoulli-design runs with median ratio 1.01 and range $[0.85, 1.13]$ (98% within $\pm 10\%$; Fig. 1c), including the 90 non-ER runs (SBM, WS, BA, Stanford3, ego-Facebook) never used for fitting; simulated leverage equals the formula within 7% for Bernoulli designs (App. B).

Cluster randomization lowers leverage; the closed form is rougher there. With $\approx$ 30–40 clusters RMSE falls from 0.22 to 0.04 (5×) on WS (d=40), from 0.120 to 0.042 (2.9×; leverage 48.9 → 5.6) on ego-Facebook and from 0.071 to 0.037 (1.9×; 47.1 → 14.6) on Stanford3, but only 1.2–1.6× on SBM and 1.0–1.3× on BA, whose clusterings cut 57%–87% of the edges. For the 126 cluster-design runs the formula underpredicts RMSE in 73% (median ratio 1.10, IQR $[1.00, 1.29]$, range $[0.69, 3.74]$); extremes occur with ≤ 10 clusters, where one draw's sample covariance of (z, s) differs from its expectation. Rank correlation between $\sqrt{L/n}$ and RMSE over all 288 runs is 0.96.

6.2 Extrapolation bias under nonlinear response does not vanish with n (H2)

Prop. 3 gives the bias of the linear one-hop regression for any response shape from exact binomial moments. Over 63 (graph, shape, degree, n) Bernoulli settings the simulated bias differs from it by at most 0.012 (median 0.003) on a GATE of ≈ 2 (Fig. 2a). It depends on the tangent slope at p: on ER (d=30) it is ≈ 0 for $k = 2$, -0.45 for $k = 4$, -0.28 for $k = \frac{1}{2}$ and $+1.47$ for the threshold response (Appendix Table 6); it grows with degree as exposures concentrate near p (Fig. 2b) and is unchanged from $n = 10^3$ to 1.6×10^4 (c). Combining the closed-form bias and leverage, $\sqrt{B^2 + \hat{\kappa}^2 L/n}$ matches the observed RMSE with median ratio 1.005 (IQR $[0.99, 1.03]$; 89% within $\pm 10\%$) over 360 world-settings (d).

Flexible regressions help only where the graph provides extreme exposures. Under Bernoulli($\frac{1}{2}$) the share of units with $s \leq 0.1$ or $s \geq 0.9$ is 0.0% on ER/regular graphs (d=30), 0.5% on BA, 6.5% on Stanford3 and 4.3% on ego-Facebook (and 79% on ego-Facebook under cluster randomization). Where such units exist, a cubic in s removes nearly all bias for every shape (Stanford3: $|\text{bias}| \leq 0.03$, RMSE 0.08–0.09 vs 0.07 for the linear model under a linear response; ego-Facebook: $|\text{bias}| \leq 0.07$, RMSE 0.17–0.18 vs 0.12), costing 15–45% RMSE when the linear model was right. On ER the cubic has RMSE 1.9–3.1 for all shapes and the quadratic leaves the bias unchanged (threshold: 1.47 → 1.47). Gradient boosting on $(z, s, s^{(2)}, \ell)$ cannot extrapolate: even for a *linear* response it is biased towards DiM (-0.52 on ER, -0.24 on Stanford3, -0.13 on ego-Facebook), but it

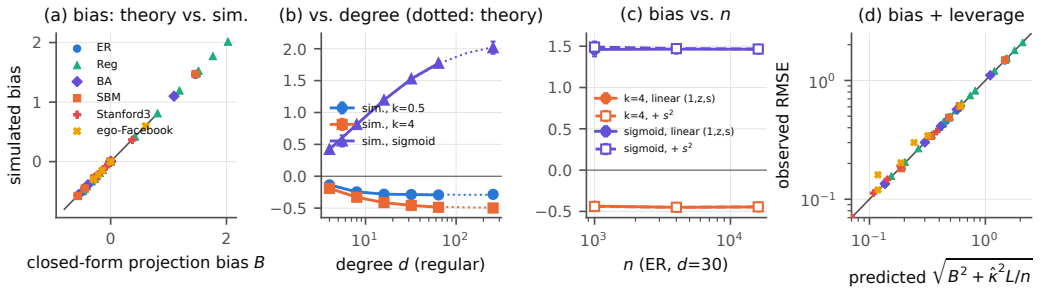


Figure 2: Extrapolation bias (E2). (a) Simulated bias of OLS-S vs the closed-form bias of Prop. 3, 63 Bernoulli settings (mean over 6 worlds). (b) Bias vs degree on random regular graphs (n=4000): simulation (solid; bars min–max) and theory (dotted). (c) Bias vs n, k=4 and threshold response, linear and $+s^2$ regressions. (d) Observed RMSE vs $\sqrt{B^2 + \hat{\kappa}^2 L/n}$ (closed-form B and L, $\hat{\kappa}$ from E1).

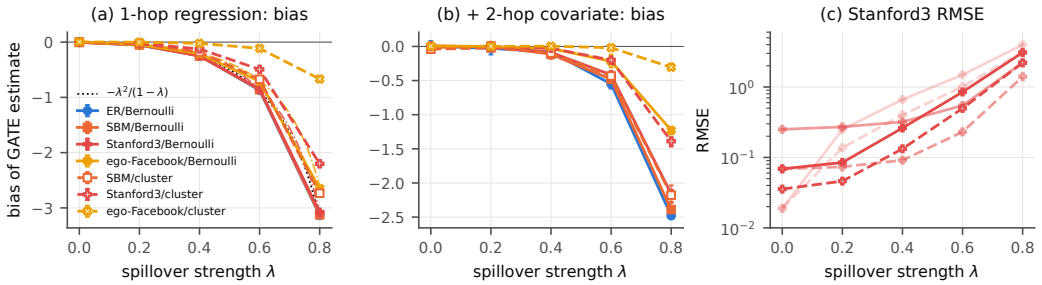


Figure 3: Multi-hop interference (E3). (a) Bias of the one-hop regression vs λ under Bernoulli (solid) and cluster designs (dashed) on four graphs; dotted black: tree-like prediction $-\lambda^2/(1-\lambda)$; dash-dotted: exact formula of Prop. 5 for ego-Facebook (yellow) and Stanford3 (red). (b) Bias after adding the two-step exposure. (c) RMSE on Stanford3: one-hop regression (dark), two-step regression (mid), DiM (light); solid Bernoulli, dashed cluster. Mean over 6 worlds, bars min–max.

has the lowest RMSE when the response is steep and the linear model badly wrong (threshold, Bernoulli: 0.52 vs 1.47 on ER, 0.26 vs 0.38 on Stanford3, 0.41 vs 0.61 on ego-Facebook) and is much worse when the linear model is adequate (0.74 vs 0.17 on ER; 3 worlds, 40 assignments). Cluster randomization shrinks the linear model's bias where clusters are good (ego-Facebook k=4: $-0.22 \to -0.03$; threshold $0.60 \to 0.05$) and little elsewhere (ER k=4: $-0.45 \to -0.36$; Stanford3: $-0.13 \to -0.12$).

6.3 Misspecified neighbourhoods, heterogeneity and exposure definition (H3, H4)

Omitted multi-hop spillover. Under the linear-in-means equilibrium the one-hop regression misses everything beyond one hop, and its bias follows Prop. 5 (Fig. 3a; Appendix Table 7): on ER (d=30) at $\lambda = 0.6$ the simulated bias is -0.872 ± 0.007 (s.e. over worlds) against -0.867 (exact walk-statistics formula) and -0.900 (tree-like approximation), a 35% relative error of a GATE of 2.49 independent of n; at $\lambda = 0.8$ it is 63%. On triangle-rich ego-Facebook the tree-like formula fails (-0.900 vs simulated -0.710) but the exact one, using the graph's return/closure statistics ψ_k, does not (-0.696); on Stanford3 both agree (-0.850 exact vs -0.846). Adding the two-step exposure $s^{(2)}$ reduces the bias (ER, λ=0.6: $-0.87 \to -0.55$; the rest comes from hops ≥ 3) but multiplies the SD by 5.4 (ER), 4.0 (Stanford3), 2.3 (ego-Facebook) under Bernoulli (λ=0.4; leverage 3560 vs 121 on ER), so its RMSE is higher than the one-hop model's in 6 of 7 graph/design combinations at $\lambda = 0.4$ and lower in 4 of 7 at $\lambda = 0.6$ (all on real graphs). **Cluster randomization absorbs part of the omitted effect:** at $\lambda = 0.6$ the bias falls from -0.71 to -0.11 on ego-Facebook, -0.85 to -0.49 on Stanford3, -0.86 to -0.67 on SBM, and grows again as clusters get finer (Fig. 4f).

Heterogeneous susceptibility. If γ_i depends on degree, OLS-S converges to a $1/d_i$-weighted average of unit effects (Prop. 4; Table 1). On Stanford3 with "inv" heterogeneity (GATE 2.0) the bias is $+2.04$ – the estimate is twice the truth – as predicted ($+2.03$, evaluated on the actual world parameters; per-

Table 1: Degree-heterogeneous spillover (E3): bias (RMSE); true GATE ≈ 2. "inv": low-degree units are more susceptible, "pos": high-degree units. B: bias predicted by Prop. 4 for OLS-S. $+c$: model with s and the neighbour count; S-DEG: s and $s\times$log-degree; WLS-S: weights $1/\mathrm{Var}(s_i)$; STRAT-S: separate fits in 5 degree strata (the last two designed after the pilot, see text). Mean over 6 worlds.

graph	γ_i het.	design	B (Prop. 4)	OLS-S	$+c$	S-DEG	WLS-S	STRAT-S
Stanford3	none	Bern.	-0.00	-0.01 (0.07)	+0.00 (0.07)	+0.01 (0.16)	+0.00 (0.19)	+0.00 (0.19)
		cluster	–	-0.00 (0.04)	+0.00 (0.04)	-0.00 (0.04)	-0.00 (0.05)	+0.00 (0.04)
	inv	Bern.	+2.03	+2.04 (2.04)	+1.55 (1.56)	-0.53 (0.56)	+0.02 (0.21)	+0.20 (0.27)
		cluster	–	+0.65 (0.70)	+0.35 (0.37)	-0.08 (0.13)	+0.21 (0.30)	+0.08 (0.09)
	pos	Bern.	-0.71	-0.71 (0.71)	-0.25 (0.26)	-0.18 (0.24)	+0.00 (0.26)	-0.04 (0.21)
		cluster	–	-0.34 (0.38)	-0.07 (0.09)	-0.05 (0.07)	-0.14 (0.24)	-0.02 (0.05)
ego-Facebook	none	Bern.	-0.00	-0.00 (0.12)	+0.01 (0.12)	+0.00 (0.19)	+0.01 (0.23)	+0.00 (0.24)
		cluster	–	+0.01 (0.04)	+0.01 (0.04)	+0.01 (0.04)	+0.00 (0.04)	+0.01 (0.05)
	inv	Bern.	+1.41	+1.41 (1.52)	+1.13 (1.22)	-0.44 (0.52)	+0.03 (0.36)	+0.18 (0.31)
		cluster	–	+0.06 (0.15)	+0.01 (0.09)	-0.01 (0.07)	+0.02 (0.15)	+0.02 (0.05)
	pos	Bern.	-0.59	-0.60 (0.65)	-0.17 (0.21)	-0.19 (0.30)	-0.05 (0.64)	-0.07 (0.30)
		cluster	–	-0.04 (0.16)	+0.01 (0.05)	-0.01 (0.07)	-0.01 (0.18)	-0.01 (0.06)

world predictions and simulations agree within 0.02); "pos": -0.71 (-0.71); ego-Facebook $+1.41$ ($+1.41$) and -0.60 (-0.59); BA graphs, with less degree spread, $+0.17$ and -0.40. Without degree-related heterogeneity it is unbiased ($|\text{bias}| \leq 0.01$). Adding the neighbour count or $s\times$log-degree reduces but does not remove the bias ($+1.55$, -0.53 on Stanford3 "inv"). Variance-weighted fitting (WLS-S) removes it ($+0.02$) but with RMSE 0.19–0.64 (vs 0.07–0.12 for OLS-S in the homogeneous case); degree stratification (STRAT-S) gives bias $+0.20$, RMSE 0.27 (vs 2.04) and is almost free under cluster randomization (RMSE 0.04–0.05). Both were devised after the pilot results.

Exposure definition. If spillover follows the neighbour *count* (scaled by mean degree) but the model uses the *fraction*, OLS-S is biased by -0.90 (Stanford3) and -0.75 (ego-Facebook); the converse mismatch gives -0.98 and -0.97 (RMSE 0.98). Including both s and c fixes both directions (RMSE 0.07/0.12 under the fraction and 0.07/0.13 under the count mechanism, vs 0.03/0.06 for the correct count model), provided the true exposure is nested in the model.

6.4 Design: treated fraction and clustering (H5)

Under Bernoulli designs RMSE scales as $(p(1-p))^{-1/2}$ (Fig. 4a): the RMSE at $p = 0.05$ is 2.2, 2.3, 2.2 times that at $p = 0.5$ on SBM, ego-Facebook, Stanford3 against a predicted 2.29, and $\hat{\kappa}\sqrt{L/n}$ matches all runs of this sweep (median ratio 0.97, range $[0.81, 1.21]$). The bias under nonlinear response also depends on p (Fig. 4b,c): with a concave response a small p gives a large positive bias ($+1.7$ on SBM at p=0.05, k=$\frac{1}{2}$) whose sign flips near $p \approx 0.3$–0.5, agreeing with Prop. 3 to a median of 0.011–0.014 (max 0.14–0.21 at $p = 0.05$, where few units are treated). Small treated fractions, typical of cautious rollouts, are thus both noisier and more extrapolation-prone.

The number of clusters trades bias against variance (Fig. 4d–f). Finer clusters cut more edges, so the bias from nonlinear response and omitted hops grows monotonically (ego-Facebook k=4: $0.00 \to -0.08$ from 7 to 559 clusters; Stanford3 multi-hop: $-0.28 \to -0.62$), while very few clusters leave few independent units (Stanford3: RMSE 0.15 with 5 clusters, one holding 87% of units, vs 0.035 with ≈ 18). For a linear response RMSE is flat from ≈ 10 (ego-Facebook; 0.04–0.05) or ≈ 18 (Stanford3; 0.035–0.045) to 600 clusters, with a shallow minimum near 80 on SBM; with a nonlinear response fewer clusters are preferred (SBM k=4: 0.34 at 40 clusters vs 0.46 at 690). With 30 clusters DiM remains biased by -0.14 (7% of the GATE) on ego-Facebook and -0.50 (25%) on Stanford3, so clustering alone does not remove the need for the regression correction.

6.5 When is regression better than the baselines? (H6)

Difference in means. In the linear one-hop model DiM estimates only the direct effect, so its bias is $-\bar{\gamma}$ (-1.00 at $\gamma = 1$ under Bernoulli on every graph), independent of n, whereas the regression is unbiased but pays leverage $L \geq 1/v$. The regression has lower RMSE than DiM only when the spillover exceeds roughly the regression's own noise floor, $\gamma^* \approx \hat{\kappa}\sqrt{L/n}$ (Fig. 5). Across nine (degree, n) combinations on ER graphs, the measured crossover (RMSE ratio $= 1$, log-interpolated,

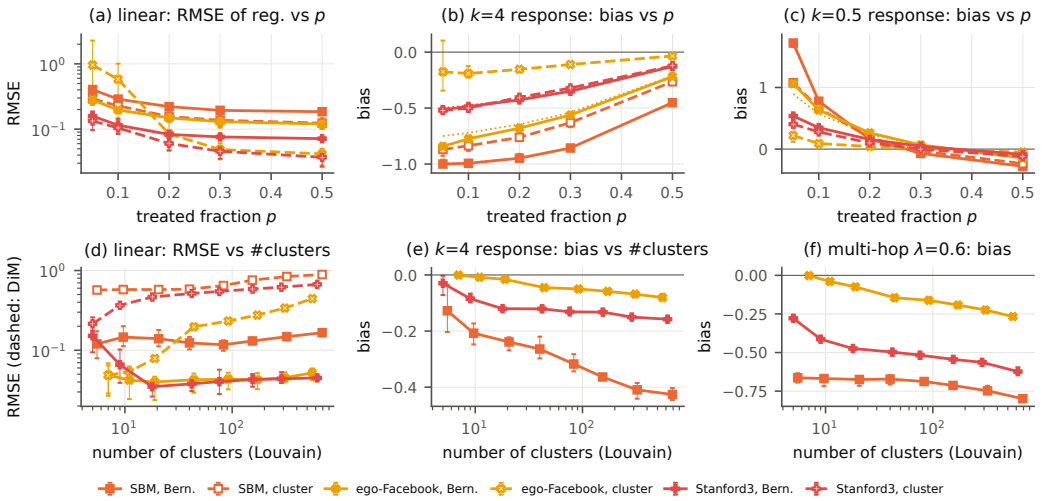


Figure 4: Design factors (E5), OLS-S estimator. Top: treated fraction p (solid Bernoulli; dashed cluster randomization with ≈30/40 clusters; dotted: closed-form bias for Bernoulli). Bottom: number of clusters (same colours/markers per graph; solid: OLS-S, dashed in (d): DiM). Mean over 6 worlds, bars min–max.

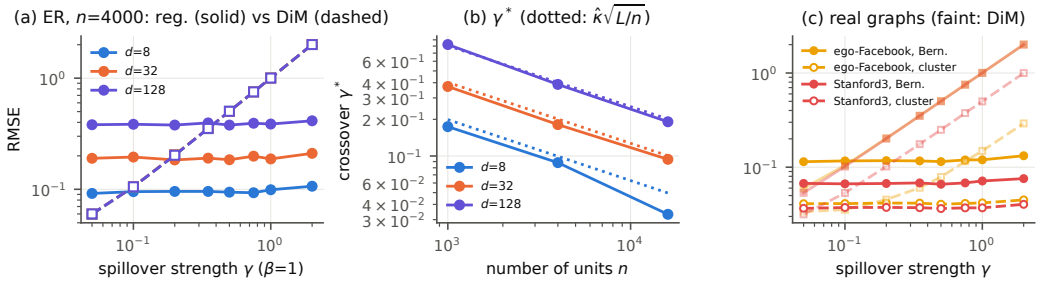


Figure 5: Regression vs DiM (E4). (a) RMSE vs γ on ER graphs (n=4000, β≈1): regression (solid) is flat at its noise floor, DiM (dashed) grows with its bias. (b) Spillover strength at which both have equal RMSE vs n for three degrees; dotted: $\hat{\kappa}\sqrt{L/n}$. (c) Real graphs, Bernoulli (solid) and clustered (dashed; hollow), regression in colour, DiM faint. Mean over 6 worlds.

mean of 6 worlds) is within 0.67–1.02× of $\hat{\kappa}\sqrt{L/n}$ (Table 5). It is large for dense graphs and small samples: with $n = 10^3$, $d = 128$ the spillover must be 0.82β before regression helps, whereas at $n = 1.6 \times 10^4$, $d = 8$ it is 0.03β. When there is no interference ($\gamma = 0$) DiM is better by a factor $\sqrt{Lv} = \sqrt{1 + d_H}$ (RMSE 0.016 vs 0.047 at $d = 8$, $n = 1.6 \times 10^4$; 0.032 vs 0.38 at $d = 128$, $n = 4000$). On the real graphs the crossover is at $\gamma \approx 0.07$ (Stanford3) and ≈ 0.12 (ego-Facebook) under Bernoulli design and at ≈ 0.06 and ≈ 0.19 under cluster randomization (Fig. 5c; read off the grid $\gamma \in \{0, 0.05, 0.1, 0.2, \dots\}$ by interpolation).

Hájek. The full-neighbourhood Hájek estimator is unbiased when defined but is undefined in 79% of draws at $d = 16$ ($n = 4000$) and 100% for $d \geq 32$ under Bernoulli (Appendix Table 5); where defined its RMSE is 1.3–1.6 on ER. On Stanford3 and ego-Facebook, whose low-degree units supply qualifying exposures, it is defined and unbiased but has RMSE 0.48 and 0.67 under Bernoulli versus 0.07 and 0.12 for the regression; with ≈ 30 clusters it is competitive on ego-Facebook (0.070 vs 0.042) and worse on Stanford3 (0.179 vs 0.037). With strongly nonlinear responses and cluster randomization on ego-Facebook it matches the linear regression (threshold: 0.07 vs 0.07; k=4: 0.07 vs 0.08), so on clusterable graphs it is a safe alternative.

6.6 Confidence intervals

We audit three standard errors for the one-hop regression (E6; $R = 400 \times 6$ worlds; 8 graph/design combinations): the HC1 sandwich over units; a sandwich clustered on Louvain communities; and

a graph-HAC allowing residuals of units within two hops to correlate [Kojevnikov et al., 2021] (coverage in Appendix Fig. 6). *Correct model:* HC coverage is near nominal (mean 0.95, range 0.87–1.00 over the 48 world-settings; mean SE/SD ratio 1.03); the clustered sandwich is slightly anticonservative (0.93, ratio 0.97) and the HAC more so (0.88, ratio 0.91; 0.79 for ER with $d = 30$), its 2-hop pattern covering 4%–34% of all pairs. *Misspecified models* (concave $k = \frac{1}{2}$, multi-hop $\lambda = 0.4$): HC coverage has mean 0.57 (range 0.06–0.83; lowest on Stanford3 where the bias is 3.2 SD). This is a bias problem: coverage predicted from the observed bias, SD and mean SE under a normal approximation matches the observed one with mean absolute difference 0.011 (correlation 0.998, 144 world-settings). The exception is clustered designs with misspecification, where the HC SE underestimates the SD (ratio 0.46 for Stanford3/concave, 0.58 for ego-Facebook/multi-hop; clustered sandwich 0.79, 0.89). No variance estimator addresses B: intervals cover the regression's population target, not the GATE.

6.7 A mis-measured network

The interference graph is itself an estimate. In E7 exposure features are computed from a graph in which a fraction f of true edges is replaced by random edges (outcomes follow the true graph, linear response). Under Bernoulli designs the one-hop regression is attenuated towards the direct effect by almost exactly $f\bar{\gamma}$ (App. Fig. 7a): bias $-0.06, -0.11, -0.22, -0.42$ for $f = 0.05, 0.1, 0.2, 0.4$ on Stanford3 (ego-Facebook, ER alike), RMSE $0.07 \to 0.45$. Under cluster randomization the degradation is milder in our runs (Stanford3: bias -0.03, RMSE $0.037 \to 0.087$ at $f = 0.4$; ego-Facebook: bias $+0.07$, RMSE $0.042 \to 0.118$); we did not investigate why, and clusters were computed on the true graph, which flatters this case. The two-step model is more fragile (bias $+0.23$ at $f = 0.05$ on Stanford3, Bernoulli).

7 Discussion and limitations

What determines the error. In our experiments the RMSE of the regression GATE estimator is well described by $\sqrt{B^2 + \hat{\kappa}^2 L/n}$ (Fig. 2d). L depends only on graph and design and can be computed *before* an experiment (Prop. 2): it grows with the harmonic-mean degree and $1/(p(1-p))$ and is cut by clusterings that keep neighbourhoods together. B is what the regression cannot represent: the response slope away from the exposures the design visits, interference beyond the modelled neighbourhood, and effect heterogeneity correlated with a unit's exposure variance. More data does not cure a nonzero B; a larger model helps only where the design or low-degree units provide support, and otherwise trades bias for leverage. Practically: (i) use L to size experiments and pick p (near $\frac{1}{2}$) and the clustering; (ii) prefer cluster randomization when the graph is clusterable, balancing bias against the number of independent units; (iii) comparing with richer specifications is a natural sensitivity check, but their leverage was up to $20\times$ larger, so only large biases are detectable (not evaluated formally); (iv) intervals inherit B.

Limitations. *Synthetic outcomes:* four stylized mechanisms with chosen parameters (γ=1, $\lambda \leq 0.8$); bias magnitudes belong to these, not to platforms, and real interference may combine them or add attribute heterogeneity and dynamics. *Known graph* except in E7 (one of many mis-measurement models). *Theory:* Props. 1–5 are first-order design-expectation statements without rates; there is no closed form for B under cluster designs, and the leverage formula underpredicts RMSE in 73% of cluster-design runs (median $+10\%$, up to $3.7\times$). *Estimators:* hand-specified exposure features, two heterogeneity fixes and one boosting model (few replications); we did not implement the covariate selection of [Han and Ugander, 2023], regularized fits or optimized clusterings (Louvain, unequal sizes). *Inference:* three variance estimators, none correcting bias. *Scale:* $\leq 1.2 \times 10^4$ nodes, 6 worlds per setting, bars are min–max not confidence intervals. *Pilot:* WLS-S/STRAT-S were designed after the heterogeneity pilot, so Table 1 is in-sample for them. Negative results (little clustering benefit on weakly clusterable graphs, no help from quadratic terms, two-step exposure raising RMSE, anticonservative HAC, biased boosting) are reported in Sec. 6.

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A Proofs

Pooled moments. Because the design is symmetric across units and we average over units, all moments below are *pooled*: $\Sigma = \frac{1}{n}\sum_i \mathbb{E}[\phi_i \phi_i^\top]$. If ϕ contains an intercept and $\bar{d}$ has a zero intercept coordinate, then $\bar{d}^\top \Sigma^{-1} \bar{d} = \tilde{d}^\top \Gamma^{-1} \tilde{d}$, where Γ is the pooled covariance matrix of the non-intercept features and $\tilde{d}$ the corresponding sub-vector of $\bar{d}$ (block inverse / Schur complement).

Proof of Proposition 1. $\hat{\beta} = \hat{\Sigma}^{-1} \frac{1}{n}\sum_i \phi_i(\phi_i^\top \beta^\star + \varepsilon_i) = \beta^\star + \hat{\Sigma}^{-1}\hat{g}$, so $\hat{\tau} - \tau = (\bar{d}^\top \beta^\star - \tau) + \bar{d}^\top \hat{\Sigma}^{-1}\hat{g}$. By definition of $\beta^\star$, $\mathbb{E}\hat{g} = \frac{1}{n}\sum_i \mathbb{E}[\phi_i y_i] - \Sigma\beta^\star = 0$. Replacing $\hat{\Sigma}$ by Σ the second term is $u^\top \Sigma^{-1/2}\hat{g}$ with $u = \Sigma^{-1/2}\bar{d}$, $\|u\|^2 = L$; its variance is $u^\top M u/n$ with $M = \Sigma^{-1/2}\, n\mathrm{Cov}(\hat{g})\, \Sigma^{-1/2}$, and $u^\top M u \le \lambda_{\max}(M)\|u\|^2 = \kappa L$ (Rayleigh quotient). The replacement of $\hat{\Sigma}$ by Σ is a first-order (delta-method) approximation; we do not prove a rate and instead test its consequence (RMSE $\approx \sqrt{B^2 + \kappa L/n}$) empirically. □

Proof of Proposition 2. Take $\phi = (1, z, s)$, $\bar{d} = (0, 1, 1)$. Since $s_i = \sum_j W_{ij} z_j$ and $\mathbb{E}z_j = p$, $\mathbb{E}s_i = p$. Under independent cluster-level Bernoulli assignment $\mathrm{Cov}(z_j, z_l) = v\,\mathbf{1}[c(j) = c(l)]$. Hence $\Gamma_{zz} = v$; $\mathrm{Cov}(z_i, s_i) = v\sum_j W_{ij}\mathbf{1}[c(j) = c(i)] = v\, w_{i,c(i)}$, so $\Gamma_{zs} = v\rho$; and $\mathrm{Var}(s_i) = v\sum_c (m_{ic}/d_i)^2$, so $\Gamma_{ss} = vq$. With $\tilde{d} = (1, 1)$, $\tilde{d}^\top \Gamma^{-1}\tilde{d} = \frac{1}{v}\frac{1-2\rho+q}{q-\rho^2} = \frac{1}{v}\big[1 + \frac{(1-\rho)^2}{q-\rho^2}\big]$. $q \ge \rho^2$ by Jensen and $\sum_c w_{ic}^2 \ge w_{i,c(i)}^2$. For unit randomization $m_{ic} \in \{0, 1\}$ for singleton clusters, $\rho = 0$, $q = \frac{1}{n}\sum_i 1/d_i = 1/d_H$. □

Proof of Proposition 3. Under Bernoulli assignment $s_i = \frac{1}{d_i}\sum_{j\sim i} z_j$ is independent of z_i (no self loops), $\mathbb{E}s_i = p$, $s_i \sim \mathrm{Bin}(d_i, p)/d_i$ and $\mathrm{Var}(s_i) = v/d_i$. Thus $\Gamma_{zs} = 0$, $\Gamma_{zz} = v$, $\Gamma_{ss} = \frac{1}{n}\sum_i v/d_i$. The fixed baselines a_i have zero pooled covariance with z and s because $\mathbb{E}z_i = \mathbb{E}s_i = p$ for all i. Therefore the pooled covariances of y with z and s are $v\bar{\beta}$ and $\gamma\frac{1}{n}\sum_i \mathrm{Cov}(g(s_i), s_i)$ (the term $\beta_i \mathrm{Cov}(z_i, s_i)$ vanishes and $\mathrm{Cov}(g(s_i), z_i) = 0$), giving the coefficients $\bar{\beta}$ and $\gamma\kappa_p$ on z and s; $\bar{d}^\top \beta^\star = \bar{\beta} + \gamma\kappa_p$ while $\tau = \bar{\beta} + \gamma(g(1) - g(0))$. For the limit, $s_i - p = O_P(d_i^{-1/2})$; for $g \in C^2$, $\mathrm{Cov}(g(s), s) = g'(p)\mathrm{Var}(s) + O(\mathbb{E}|s - p|^3) = g'(p)\mathrm{Var}(s) + O(d^{-3/2})$ hence $\kappa_p \to g'(p)$ as $\min_i d_i \to \infty$. (For $g(s) = s^2$ and $p = \frac{1}{2}$ one finds $\kappa_p = 1$ exactly for every d.) The quantity κ_p is evaluated by exact binomial sums for each observed degree in our code (`theory.py`). □

Proof of Proposition 4. As above $\Gamma_{zz} = v$, $\Gamma_{zs} = 0$, $\Gamma_{ss} = v\frac{1}{n}\sum_i d_i^{-1}$. Pooled $\mathrm{Cov}(y, z) = v\bar{\beta}$ and $\mathrm{Cov}(y, s) = \frac{1}{n}\sum_i \gamma_i \mathrm{Var}(s_i) = \frac{v}{n}\sum_i \gamma_i d_i^{-1}$. The coefficient on s is $\sum_i \gamma_i d_i^{-1} / \sum_j d_j^{-1} = \sum_i \pi_i \gamma_i$. □

Proof of Proposition 5. $Y(z) = \sum_{k\ge 0} \lambda^k W^k (a + \beta z)$ (Neumann series). Under Bernoulli assignment $\mathrm{Cov}(z_i, (W^k z)_i) = v(W^k)_{ii}$ and $\mathrm{Cov}(s_i, (W^k z)_i) = v\sum_j W_{ij}(W^k)_{ij}$. Pooling over i and using $\Gamma_{zs} = 0$, $\Gamma_{zz} = v$, $\Gamma_{ss} = vq$: the coefficient on z is $\beta\sum_{k\ge 0}\lambda^k r_k = \beta(1 + \sum_{k\ge 2}\lambda^k r_k)$ ($r_0 = 1$, $r_1 = 0$) and the coefficient on s is $\beta\sum_{k\ge 1}\lambda^k u_k/q = \beta(\lambda + \sum_{k\ge 2}\lambda^k u_k/q)$. Their sum minus $\tau = \beta\sum_{k\ge 0}\lambda^k$ is $-\beta\sum_{k\ge 2}\lambda^k(1 - \psi_k)$. On locally tree-like, high-degree graphs $r_k, u_k/q \to 0$ for $k \ge 2$ and $B \to -\beta\lambda^2/(1 - \lambda)$. □

B Additional results

Notes. (i) In Table 2 the closed-form leverage is essentially exact for Bernoulli designs (within 7% in every run, within 2% in the median family) and underestimates the simulated leverage for cluster designs, by a median of 3–40% per family and by a factor up to 15.6 (Stanford3 with 9 clusters), consistent with Prop. 2 being an expectation over assignments. (ii) In Table 4 the closed-form leverage is accurate once there are about ≥ 20 clusters. (iii) Table 3: RMSE scales sublinearly in σ_a because effect heterogeneity (coefficients β_i, γ_i with 50% relative SD) adds a baseline-independent component.

Table 2: Closed-form leverage vs simulation (E1, OLS-S). Runs are (setting, world) pairs. Left: simulated mean leverage over R assignments divided by L of Prop. 2; right: observed RMSE divided by $\hat{\kappa}\sqrt{L/n}$ (median [min, max]).

family/design	runs	$\bar{L}_{\mathrm{sim}}/L$	RMSE / predicted
BA/Bern.	24	1.01 [1.00, 1.02]	1.01 [0.91, 1.05]
BA/cluster	24	1.40 [1.10, 2.33]	1.23 [0.90, 1.66]
ER/Bern.	72	1.01 [1.00, 1.07]	1.01 [0.89, 1.10]
SBM/Bern.	30	1.01 [1.00, 1.04]	1.00 [0.92, 1.10]
SBM/cluster	30	1.36 [1.12, 2.65]	1.20 [0.85, 1.75]
Stanford3/Bern.	6	1.00 [1.00, 1.00]	1.03 [0.98, 1.09]
Stanford3/cluster	24	1.15 [1.05, 15.57]	1.11 [0.77, 3.74]
WS/Bern.	24	1.01 [1.00, 1.03]	1.00 [0.91, 1.08]
WS/cluster	24	1.03 [1.03, 1.04]	1.00 [0.85, 1.19]
ego-Facebook/Bern.	6	1.02 [1.02, 1.02]	1.05 [0.85, 1.13]
ego-Facebook/cluster	24	1.07 [1.03, 1.30]	1.08 [0.69, 1.50]

Table 3: Dependence on the baseline scale σ_a (E1; ER, $n = 4000$, $d = 16$, Bernoulli; effect heterogeneity fixed at 0.5).

σ_a	RMSE	bias
0.25	0.065	-0.002
0.50	0.085	+0.004
1.00	0.132	+0.003
2.00	0.254	-0.004
4.00	0.503	-0.002

C Reproducibility details

Code and instructions. All code, raw results (CSV), logs and this paper's source are in the released archive; `README.md` gives the full procedure and `run_all_experiments.sh` runs it end to end. **Environment:** Python 3.11.2; numpy 2.4.6, scipy 1.17.1, networkx 3.6.1, pandas 3.1.1, scikit-learn 1.9.1, matplotlib 3.11.2 (`requirements.txt`); pdfLaTeX (TeX Live) with the official `neurips_2026.sty`. **Hardware/time:** CPU only; the container was limited to 4 CPU cores (cgroup quota) with 251 GB RAM available (memory use was not profiled; each worker held at most one graph and, for E6's HAC, one $\approx 10^7$–10^8-entry sparse matrix). Wall-clock times of the final runs on 4 processes: E1 172 s, E2 473 s, E2g (boosting) 181 s, E3 355 s, E4 177 s, E5 1269 s, E6 858 s, E7 64 s, exact walk statistics (`psi.py`) ≈ 4 min; about 1 h in total (≈ 4 CPU-hours); the 3-world pilot of E1/E3 cost a few minutes. **Data.** Stanford3: `https://nrvis.com/download/data/socfb/socfb-Stanford3.zip` (Network Repository, CC BY-SA; original source Traud et al. [Traud et al., 2012]; file `socfb-Stanford3.mtx`, 11586×11586, 568,309 edges; zip SHA-256 `b1221e41...e8720d`, verified, 1,649,740 bytes). ego-Facebook: `https://snap.stanford.edu/data/facebook_combined.txt.gz` (SNAP; McAuley and Leskovec [McAuley and Leskovec, 2012]; 88,234 edges, 4,039 nodes; SHA-256 `125e84db...35d7`, verified, 218,576 bytes). Downloads are parsed as plain text only (never executed) by `code/graphs.py`; both graphs are connected so the largest-component restriction is the identity. Synthetic graphs use `networkx`. **Seeds.** Synthetic graph of world i: seed $100 + i$; outcome parameters of world i: seed $1000 + i$ ($i = 0..5$); the assignment RNG of each configuration is seeded by the first 8 hex digits of the SHA-256 of its (graph, mechanism, design, i) description (`experiments.stable_seed`); Louvain clustering uses seed 1 (cached on disk for the real graphs). We re-ran E7 and obtained bit-identical numeric output (the other experiments use the same mechanism; they were not re-run end to end). **Script → output.** Fig. 1 `fig_variance.py` (E1); Fig. 2 `fig_extrap.py` (E2; $\hat{\kappa}$ from `stats_summary.py`); Fig. 3 `fig_multihop.py` (E3, `psi.py`); Fig. 4 `fig_crossover.py` (E4); Fig. 5 `fig_design.py` (E5); Figs. 6–7 `fig_inference.py`, `fig_obsnoise.py` (E6, E7); Table 1 `mk_tables.py:tab_het` (E3); Tables 2–7 `mk_tables.py` (E1–E5); every number quoted in the text is printed by `stats_summary.py` or `verify_claims.py`. **Literature access.** Only the starter literature and unrelated public papers were consulted (searches by title for the cited works). No page matching the withheld paper was knowingly opened; a few search-result titles/abstract snippets of unrelated network-interference papers were seen incidentally and none was read further.

Table 4: Clustering quality and error (E5, linear response, OLS-S, $p = \frac{1}{2}$): number of Louvain clusters, fraction of edges cut, simulated mean leverage, closed-form leverage, RMSE (means over 6 worlds).

graph	clusters	cut	$\bar{L}_{\text{sim}}$	L	RMSE
SBM	6	0.44	72.3	13.7	0.120
SBM	10	0.50	61.6	21.5	0.146
SBM	21	0.55	55.5	31.7	0.140
SBM	40	0.57	54.4	40.5	0.124
SBM	83	0.63	43.5	35.9	0.117
SBM	154	0.75	51.8	45.7	0.131
SBM	324	0.84	72.5	67.3	0.148
SBM	690	0.89	91.1	87.5	0.166
Stanford3	5	0.02	461.3	4.3	0.151
Stanford3	9	0.34	56.2	7.3	0.066
Stanford3	18	0.53	14.1	10.4	0.035
Stanford3	42	0.61	14.8	12.6	0.038
Stanford3	76	0.67	15.4	13.9	0.040
Stanford3	155	0.72	16.5	15.4	0.043
Stanford3	288	0.76	17.6	16.7	0.044
Stanford3	621	0.81	20.4	19.8	0.045
ego-Facebook	7	0.00	6.6	4.1	0.049
ego-Facebook	11	0.03	5.5	4.3	0.042
ego-Facebook	19	0.06	5.3	4.6	0.040
ego-Facebook	44	0.32	5.9	5.5	0.043
ego-Facebook	92	0.41	6.2	5.8	0.043
ego-Facebook	172	0.49	6.7	6.4	0.043
ego-Facebook	310	0.58	7.4	7.1	0.045
ego-Facebook	559	0.70	9.0	8.7	0.052

Table 5: Crossover spillover strength γ^* above which the one-hop regression has lower RMSE than DiM (E4; ER graphs, Bernoulli; $\beta \approx 1$), versus $\hat{\kappa}\sqrt{L/n}$; RMSE of DiM and regression at $\gamma = 0$. Hájek (E1, $n = 4000$) is undefined in a fraction 0.00, 0.00, 0.79, 1.00, 1.00, 1.00, 1.00 of draws for $d = 4, 8, 16, 32, 64, 128, 256$ respectively.

d	n	γ^* (obs.)	$\hat{\kappa}\sqrt{L/n}$	DiM RMSE	reg. RMSE
8	1000	0.174	0.199	0.064	0.190
8	4000	0.088	0.099	0.033	0.095
8	16000	0.033	0.050	0.016	0.047
32	1000	0.372	0.402	0.064	0.376
32	4000	0.181	0.201	0.032	0.191
32	16000	0.094	0.100	0.017	0.094
128	1000	0.823	0.805	0.067	0.822
128	4000	0.387	0.402	0.032	0.380
128	16000	0.191	0.201	0.016	0.187

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Table 6: Nonlinear exposure response (E2): bias (RMSE) of the GATE estimate; true GATE ≈ 2. Mean over 6 worlds (R=300). † GBM: 3 worlds, R=40 assignments each. "–": Hájek undefined in $\geq 50\%$ of draws (or not run).

graph	response	design	OLS-S	$+s^2$	$+s^2, s^3$	GBM†	Hájek
ER	k=4	Bern.	-0.45 (0.50)	-0.45 (0.50)	+0.01 (1.92)	-0.65 (0.83)	–
		cluster	-0.36 (0.52)	-0.45 (0.54)	+0.03 (1.51)	-0.60 (0.72)	–
	threshold	Bern.	+1.46 (1.48)	+1.47 (1.49)	-2.34 (3.06)	-0.14 (0.52)	–
		cluster	+0.94 (1.02)	+1.48 (1.52)	-0.65 (2.27)	-0.08 (0.41)	–
ego-Facebook	k=4	Bern.	-0.22 (0.30)	-0.24 (0.27)	-0.01 (0.17)	-0.17 (0.49)	-0.01 (0.65)
		cluster	-0.03 (0.08)	-0.03 (0.06)	+0.00 (0.05)	-0.01 (0.22)	-0.00 (0.07)
	threshold	Bern.	+0.60 (0.61)	+0.64 (0.66)	-0.07 (0.18)	+0.12 (0.41)	-0.01 (0.63)
		cluster	+0.05 (0.07)	+0.05 (0.07)	-0.01 (0.05)	-0.01 (0.25)	-0.00 (0.07)
Stanford3	k=4	Bern.	-0.13 (0.15)	-0.13 (0.15)	-0.01 (0.08)	-0.26 (0.37)	-0.01 (0.50)
		cluster	-0.12 (0.21)	-0.13 (0.14)	-0.00 (0.06)	-0.11 (0.21)	-0.00 (0.17)
	threshold	Bern.	+0.37 (0.37)	+0.37 (0.37)	-0.03 (0.09)	-0.11 (0.26)	-0.03 (0.48)
		cluster	+0.20 (0.20)	+0.23 (0.24)	-0.06 (0.08)	-0.06 (0.14)	+0.00 (0.16)

Table 7: Multi-hop spillover, $\lambda = 0.6$ (E3): bias (RMSE) of DiM, one-hop regression and regression with two-step exposure; last two columns: closed-form bias of the one-hop regression under Bernoulli designs (exact walk-statistics formula, tree-like approximation). Mean over 6 worlds.

graph/design	GATE	DiM	OLS-S	OLS-S2	B exact	B tree
ER/Bern.	2.49	-1.48 (1.48)	-0.87 (0.89)	-0.55 (1.19)	-0.867	-0.900
SBM/Bern.	2.49	-1.49 (1.49)	-0.86 (0.88)	-0.48 (1.16)	–	-0.900
SBM/cluster	2.49	-1.15 (1.15)	-0.67 (0.69)	-0.43 (0.83)	–	–
ego-FB/Bern.	2.49	-1.47 (1.47)	-0.71 (0.76)	-0.22 (0.65)	-0.696	-0.900
ego-FB/cluster	2.49	-0.34 (0.35)	-0.11 (0.16)	-0.02 (0.13)	–	–
Stanford3/Bern.	2.50	-1.49 (1.49)	-0.85 (0.85)	-0.42 (0.55)	-0.850	-0.900
Stanford3/cluster	2.50	-1.01 (1.01)	-0.49 (0.50)	-0.20 (0.23)	–	–

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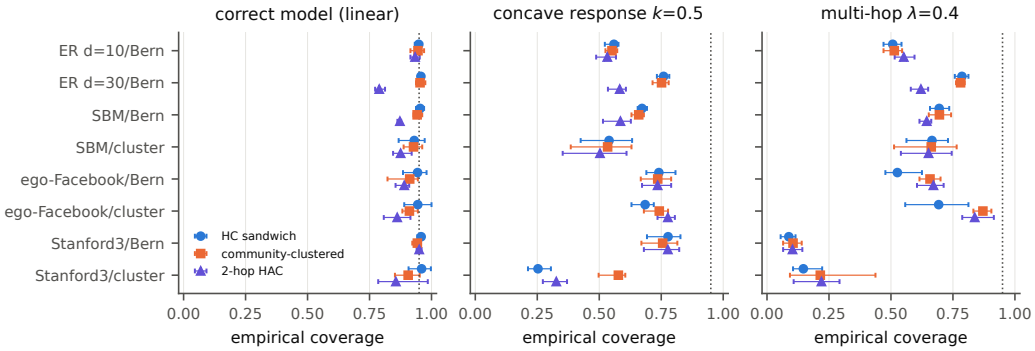


Figure 6: Coverage of nominal 95% intervals for the one-hop regression estimator (E6). Markers: mean over 6 worlds; bars min–max. Dotted: 0.95.

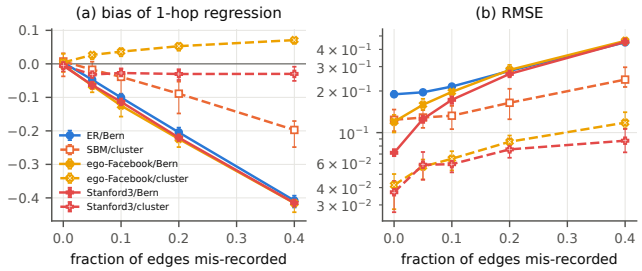


Figure 7: Mis-measured network (E7): (a) bias and (b) RMSE of the one-hop regression vs fraction of edges replaced by random edges (mean over 6 worlds; bars min–max).

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

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